



THE UNIVERSITY OF
TENNESSEE
KNOXVILLE

EPP 575 Introduction to RNA-Seq, Summer 2026

University of Tennessee, Knoxville

Course Section: CRN87151

Meeting Time and Place:

June 3-5 and June 8-10

9:00am to 11:30am

The Agriculture & Natural Resources Building (ANRB) 124

Course Credit Hours: 1

Faculty Contact Information

Primary Instructors:

Ryan Kuster (rkuster@utk.edu) Office ANRB 353

Office hours are daily 12:00pm – 1:00pm (ANRB 353) or can be scheduled by email.

Course website: https://github.com/ryandkuster/EPP_575_RNA_26/wiki

Course Description/Information:

This one-credit course will cover the basics of using a command line interface on UT's Linux-based computational resources to analyze RNA-Seq data. Basic steps such as quality assessment, read mapping and differential gene expression statistical analysis will be covered. This class meets in ANRB 124.

Value Proposition:

The demand for bioinformatics skills has become a prerequisite in many modern research approaches in the biological sciences. This course will provide students with an introduction to key bioinformatic skills, including familiarity with UNIX command line, common R programming packages and use of software for processing common bioinformatics data types in relation to bulk RNA-Seq analysis.

Student Learning Outcomes/Objectives:

1. Students will be able to apply basic bioinformatic theory and tools to analyze transcriptome datasets
2. Students will be able to effectively communicate and critically assess the application of bioinformatic tools to transcriptome data
3. Students will have a basic competence of the UNIX shell and usage of bioinformatic tools from the command line

Learning Environment:

Class meets on the days listed from 9:00am to 11:30am in ANRB 124. Classes will have a combination of lectures, guided computer labs, and short group/class discussions.

A classroom is a collaborative environment, and both the instructor and the students have a shared responsibility to ensure a successful learning experience. Students are encouraged to ask questions regarding the lectures, labs, and applications of skills in a research-oriented mindset.

Course Communications:

Outside of class and the website, the instructors will utilize email to communicate course information, such as changes to the syllabus, answering questions relevant to all students, etc. All students are responsible for checking their university email accounts and reading all emails regarding the class.

Students are welcome to drop in for office hours (12:00pm-1:00pm after class) or email to schedule an appointment.

For technical issues, contact the OIT HelpDesk via phone (865) 974-9900 or online at <http://help.utk.edu/>.

How to Be Successful in This Course:

- Follow along with the exercises during the class period. The concepts and practical exercises build on the material covered in prior lessons, so it is essential to keep up with the subject matter.
- Get help early with problems. The instructors are there to help and want you to be successful. If something is not making sense or you are unable to complete a lab exercise, seek help immediately through email and/or in-person meetings. This will prevent you from falling behind during this fast-paced class.
- Begin thinking of a scientific question for the final project and work on it before the class ends. The scope of the project should be informed by the RNA-Seq processing and analytical steps we address in class but should also include an aspect of the data or methodology you find interesting. **If you aren't sure, you can email your planned analysis to the instructor to see if it is of proper scope.**

Texts/Resources/Materials:

The course website will be used to distribute reading materials, links to references, lecture slides, and laboratory exercises. There is not a required textbook to purchase.

Required Equipment:

Students are required to bring their own laptops (and power cord if needed) to class.

Course Resources:

The course website will be used to distribute reading materials, links to references, lecture slides, and laboratory exercises. There is not a required textbook to purchase.

Course Requirements, Assessments, and Evaluations:

Presence during lecture and lab is beneficial for students to develop an understanding of the material presented in the class. Absences due to special circumstances should be discussed with the instructor, preferably prior to the absence, via email or in person. Lectures and labs will be recorded and can be viewed remotely.

The final grade for each student will be on an A-F scale:

A	93-100 points
B+	88-92 points
B	80-87 points
C+	77-79
C	70-76
F	below 70

Points will be accrued from three assignments, with the following weight:

- Daily email of output files from lab exercises and questions (posted per day on the course website) (**12 points per day, up to 60 points total for class**)
- Final project (**up to 40 points**)

Grading details:

Daily output files : At the end of each day, you will need to send a message and/or files via email to rkuster@utk.edu to get credit for completing the lab exercises. That exercise information will be specified through the course website on the daily agenda page.

Final Project : Each student will be assigned a randomized subset of replicates from the study Ding et al. (2022) dataset from the course. Submit a zipped folder containing *reproducible* documentation. This can be in any desired format, preferably markdown or similar, and must include all command line and R steps used in the analysis. Please include all relevant figures. The project is due the following **Wednesday, June 24th. Email to rkuster@utk.edu.**

Upstream Processing & QC (5 pts)

- Run QC on your raw FASTQ files and summarize trends per sample and in general regarding data quality.
- Align reads to TAIR10 using STAR; report alignment statistics per sample and any trends.
- Quantify reads with featureCounts.
- Interpret your QC: list any problematic samples and describe what seems to be wrong and how it might translate to downstream analyses.

Differential Expression Analysis (10 pts)

- Import counts into DESeq2; document your experimental design formula and justify it
- Generate a PCA plot and sample distance heatmap; interpret the clustering.
- Run DESeq2 with appropriate contrasts; filter at $\text{padj} < 0.05$ and $|\log_2\text{FC}| \geq 1$.
- Produce a volcano plot and/or MA plot with notable genes annotated.
- Check expression of CPK28, NLP7, CBF1–3, and at least three COR genes (e.g., COR15A, COR47, KIN1) and visualize them in a heatmap or similar plot.

Exploratory Analysis (10 pts)

- Choose one way to investigate your dataset using an approach we did not discuss in class (please feel welcome to discuss with the instructor in advance if you feel uncertain). Possibilities include but are not limited to:
 - GO enrichment on your DEG list (clusterProfiler or topGO).
 - Replicate downsampling to assess statistical power.
 - DEG overlap across conditions using Venn or UpSet plot.
 - Use pseudo-counting tools like Salmon/Kallisto and compare to STAR/featurecounts results.
 - Anything else that you find interesting.
- Clearly state your question, methods, and findings.

Written Narrative (15 pts)

- Discuss whether your dataset's results support or contradict the Ding et al. paper's conclusions, or if you discover something else entirely.
- Describe one thing that didn't work as expected and how you handled it.
- Reflect on the your exploratory analysis and the analytical/biological interpretations.
- Target: 400–600 words, written in manuscript Results/Discussion style

Course Feedback:

A final course evaluation will be provided to each student at the end of the course. Each student will receive an email toward the end of the course providing a link to the survey. **Student feedback is vital to further developing and improving the course.**

Key Campus Resources for Students:

- [Center for Career Development and Academic Exploration](#) (Career counseling and resources; Handshake job search system)

- [Course Catalogs](#) (Listing of academic programs, courses, and policies)
- [Hilltopics](#) (Campus and academic policies, procedures and standards of conduct)
- [OIT HelpDesk](#) (865) 974-9900
- [Schedule of Classes/Timetable](#)
- [Student Health Center](#) (visit the site for a list of services)
- [Academic Success Center](#) (Academic support resources)
- [Undergraduate Academic Advising](#) (Advising resources, course requirements, and major guides)
- [University Libraries](#) (Access to library resources, databases, course reserves, and services)

Course Outline/Assignments/Units of Instruction/Clinic Schedule:

The Campus Syllabus

Dear Student,

The purpose of this **Campus Syllabus** is to provide you with important information that applies to all UTK courses. Please observe the following policies and familiarize yourself with the university resources listed below. At UT, we are committed to providing you with a high-quality learning experience. I want to wish you the best for a successful and productive semester.

—Dr. John Zomchick, Provost and Senior Vice Chancellor

ACADEMIC INTEGRITY

Each student is responsible for their personal integrity in academic life and for adhering to UT's Honor Statement. The Honor Statement reads: "An essential feature of the University of Tennessee, Knoxville is a commitment to maintaining an atmosphere of intellectual integrity and academic honesty. As a student of the university, I pledge that I will neither knowingly give nor receive any inappropriate assistance in academic work, thus affirming my own personal commitment to honor and integrity."

YOUR ROLE IN IMPROVING THE COURSE THROUGH ASSESSMENT

At UT, it is our collective responsibility to improve the state of teaching and learning. During the semester you may be requested to assess aspects of this course, either during class or at the completion of the class, and through the TNVoice course evaluation. Please take the few moments needed to respond to these requests as they are used by instructors, department heads, deans and others to improve the quality of your UT learning experience.

STUDENTS WITH DISABILITIES — <http://sds.utk.edu>

The University of Tennessee, Knoxville, is committed to providing an inclusive learning environment for all students. If you anticipate or experience a barrier in this course due to

a chronic health condition, a learning, hearing, neurological, mental health, vision, physical, or other kind of disability, or a temporary injury, you are encouraged to contact [Student Disability Services](#) (SDS) at 865-974-6087 or sds@utk.edu. An SDS Coordinator will meet with you to develop a plan to ensure you have equitable access to this course. If you are already registered with SDS, please contact your instructor to discuss implementing accommodations included in your course access letter.

ACCESSIBILITY POLICY AND TRAINING – <http://accessibility.utk.edu>

The University of Tennessee, Knoxville, provides reasonable accommodations for individual students with disabilities through its office of [Student Disability Services](#). The university is also committed to making information and materials accessible, when possible. Resources and assistance to support these efforts can be found at <http://accessibility.utk.edu/>.

WELLNESS – <http://wellness.utk.edu/> and <http://counselingcenter.utk.edu/>

The [Center for Health Education and Wellness](#) empowers all Volunteers to thrive by cultivating personal and community well-being. The Center can answer questions about general wellness, substance use, sexual health, healthy relationships, and sexual assault prevention. The [Student Counseling Center](#) is the university's primary facility for personal counseling, psychotherapy, and psychological outreach and consultation services.

[The Student Health Center](#) has a wide range of clinics and services. Their goal is to provide quality healthcare and assist you in maintaining a healthy lifestyle to support your success as a student. To schedule an appointment, call 865-974-5080. To report a health-related absence visit tiny.utk.edu/absence.

Any student who has difficulty affording hygiene products, groceries, or accessing sufficient food to eat every day is urged to contact [Basic Needs](#) for support. The Big Orange Pantry, located on the 3rd floor of the Student Union, is a free resource for all students, faculty, and staff, no matter how great or small their need is. Students who need emergency financial assistance can also request funding from the [Student Emergency Fund](#).

Students who are experiencing non-academic difficulty or distress and need assistance should call 974-HELP or [submit an online referral](#). The 974-HELP team specializes in aligning resources and support to students experiencing mental health distress.

EMERGENCY ALERT SYSTEM – <https://prepare.utk.edu/be-ready>

The University of Tennessee is committed to providing a safe environment to learn and work. When you are alerted to an emergency, please take appropriate action. Learn more about what to do in an emergency and sign up for [UT Alerts](#). Check the emergency posters near exits and elevators for building specific information. In the event of an emergency, the course schedule and assignments may be subject to change. If changes to graded activities are required, reasonable adjustments will be made, and you will be responsible for meeting revised deadlines.

